Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

</tool_call> reasoning process here
The de Broglie wavelength Î» of a particle is given by:

Î» = h / p

where:
- h = Planck's constant = 6.626 ÃĹ 10âģ»Â³âģ´ J Â·s = 6.626 ÃĹ 10âģ»Â²âģ· erg Â·s
- p is the momentum of the particle

For a non-relativistic electron with kinetic energy K, the kinetic energy is:

K = pÂ² / (2m)

So,

p = âĪļ(2mK)

Thus,

Î» = h / âĪļ(2mK)

We need to compute the de Broglie wavelength for an electron with kinetic energy K = 1 eV.

First, convert 1 eV to joules:

1 eV = 1.602 ÃĹ 10âģ»Â¹âģ¹ J

Mass of electron m = 9.109 ÃĹ 10âģ»Â³Â¹ kg

So,

p = âĪļ(2 ÃĹ 9.109 ÃĹ 10âģ»Â³Â¹ ÃĹ 1.602 ÃĹ 10âģ»Â¹âģ¹)
= âĪļ(2 ÃĹ 9.109 ÃĹ 1.602 ÃĹ 10âģ»âģµâģ°)
âīĪ âĪļ(2.92 ÃĹ 10âģ»âģ´âģ¹)
âīĪ 1.71 ÃĹ 10âģ»Â²âģµ kg Â·m/s

Now,

Î» = h / p = (6.626 ÃĹ 10âģ»Â³âģ´) / (1.71 ÃĹ 10âģ»Â²âģµ)
âīĪ 3.87 ÃĹ 10âģ»âģ¹ m

Convert to cm: 3.87 ÃĹ 10âģ»âģ¹ m = 3.87 ÃĹ 10âģ»âģ· cm

Now compare to the options:

A. 2.48 ÃĹ 10âģ»âģ¸ cm âĨĴ 2.48 ÃĹ 10âģ»âģ¹ m âĨĴ too small
B. 5.92 ÃĹ 10âģ· cm/sec âĨĴ unit is wrong (velocity)
C. 8.66 ÃĹ 10âģ»âģ¶ cm âĨĴ 8.66 ÃĹ 10âģ»âģ¸ m âĨĴ too small
D. 3.14 ÃĹ 10âģ»âģµ cm âĨĴ 3.14 ÃĹ 10âģ»âģ· m âĨĴ close but low
E. 9.81 ÃĹ 10âģ»âģ´ cm âĨĴ ~10âģ»âģµ m âĨĴ wrong
F. 5.27 ÃĹ 10âģ»âģ¹ m âĨĴ 5.27 ÃĹ 10âģ»Â¹âģ° cm âĨĴ not matching
G. 1.23 ÃĹ 10âģ»âģ· cm âĨĴ 1.23 ÃĹ 10âģ»âģ¹ m âĨĴ too small
H. 4.14 ÃĹ 10âģ»Â¹âģ° m âĨĴ 4.14 ÃĹ 10âģ»Â¹Â¹ cm âĨĴ smaller than calculated
I. 6.625 ÃĹ 10âģ»Â²âģ· erg sec âĨĴ unit is h, not wavelength
J. 0.512 ÃĹ 10âģ¶ eV âĨĴ energy, not wavelength

Now verify with exact known formula:

For electrons, a handy approximation is:

Î» (in nm) = 1.226 / âĪļ(K in eV)

So for K = 1 eV,

Î» = 1.226 / âĪļ1 = 1.226 nm = 1.226 ÃĹ 10âģ»âģ¹ m = 1.226 ÃĹ 10âģ»Â¹Â¹ cm

Wait âĢĶ something seems off.

Wait: the standard formula is:

Î» (in nanometers) = 1.22